

Multi-Card Retrieval: Purpose-Specific Embedding Views for Aspect-Rich Corpora

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Abstract. A document that covers several distinct matters is usually indexed as one dense vector, which averages those matters together. We quantify the resulting loss and show how to avoid it cheaply. On a controlled corpus in which each document carries exactly k near-orthogonal aspects, the similarity between an aspect-directed query and a pooled document vector decays as $k^{-0.32}$ while the maximum over per-aspect embeddings stays flat, and retrieval quality follows: at $k = 10$, nDCG@10 falls from 0.682 to 0.273 under pooling but holds at 0.672 under per-aspect cards. On the LIMIT capacity benchmark, where single-vector retrieval is known to fail, the decisive comparison is against chunking rather than against pooling: splitting the same text into fixed windows lifts nDCG@10 by 0.076, while splitting it into purpose-aligned cards lifts it by 0.674. The benefit therefore comes from aligning the unit of representation with the unit the query asks about, not from storing more vectors per document. We report the storage multiplier each corpus actually incurs, and we release the code, the seeds, and every per-query score.

Keywords: Dense retrieval · Multi-vector representations · Aspect-based retrieval · Reproducibility

1 Introduction

Enterprise corpora are full of documents that are about several things at once.

Contributions.

1. A measurement of aspect dilution on a controlled corpus, giving a fitted decay exponent that is milder than the orthogonality argument predicts (Section 3).
2. Evidence that purpose alignment, not embedding count, drives the benefit, via a chunking control on identical source text (Section 4).
3. An open, seeded, single-command reproduction of every number reported.

2 Related work

3 Aspect dilution

We first measure the effect in isolation, on a corpus built so that the number of aspects per document is a controlled variable.

Table 1. Aspect dilution. Deltas are per-query paired differences, card system minus pooled, with 95% bootstrap intervals and paired permutation p -values Holm-corrected across all 21 comparisons. Relevance-set sizes grow with k by construction, so nDCG is comparable down a column and between the two systems within a row, but not across rows.

	k	pooled	cards	Δ	95% CI	win/tie/loss
1	0.682	0.682	+0.000		exact tie	0/200/0
2	0.395	0.571	+0.177	[+0.145, +0.207]		160/0/40
3	0.308	0.565	+0.257	[+0.226, +0.288]		173/1/26
4	0.271	0.556	+0.285	[+0.253, +0.317]		175/0/25
5	0.300	0.585	+0.285	[+0.251, +0.318]		176/2/22
7	0.282	0.656	+0.374	[+0.339, +0.408]		182/0/18
10	0.273	0.672	+0.399	[+0.360, +0.438]		186/1/13

Construction. Each document concatenates k passages drawn from k of ten mutually unrelated topic pools. Every passage realises one *fact*, a (pool, attribute, action) triple, and a query asks about one fact in wording that shares no template with the passages. Relevance is therefore defined by the generative process rather than assumed: every document carrying a passage that realises the queried fact is relevant, which yields relevance sets of 3.7 documents per query at $k = 1$ rising to 33.3 at $k = 10$. Two representations of the same document are compared: one pooled embedding of the whole text, and k per-aspect embeddings scored by their maximum. Both are scored at document level. We use 500 documents and 200 queries per setting, encoder `all-MiniLM-L6-v2`, seed 13.

Result. Table 1 and Figure 1 report the outcome. The per-aspect representation holds its similarity to relevant documents almost exactly constant as k grows, from 0.498 at $k = 1$ to 0.493 at $k = 10$, while the pooled representation decays monotonically to 0.235. Retrieval follows: the pooled system loses more than half its nDCG@10 between $k = 1$ and $k = 2$ and does not recover, while the card system stays between 0.556 and 0.682 throughout. At $k = 1$ the two systems are identical by construction, and the measurement confirms it exactly, which is a useful check on the pipeline.

The theory has the right shape and the wrong magnitude. Treating a pooled vector as the normalised mean of k near-orthogonal aspect vectors predicts that an aspect-directed query’s similarity falls as $1/\sqrt{k}$. The measured ratio of pooled to card similarity instead follows $k^{-0.32}$, well above the prediction at every k . Two properties of real encoders explain the gap, and both work in the same direction: aspects drawn from natural language are not orthogonal, since all of the passages share register and syntax, and encoder similarities are anisotropic, occupying a narrow cone that compresses differences. We therefore report the orthogonality argument as an upper bound on dilution rather than as a prediction that the data confirm.

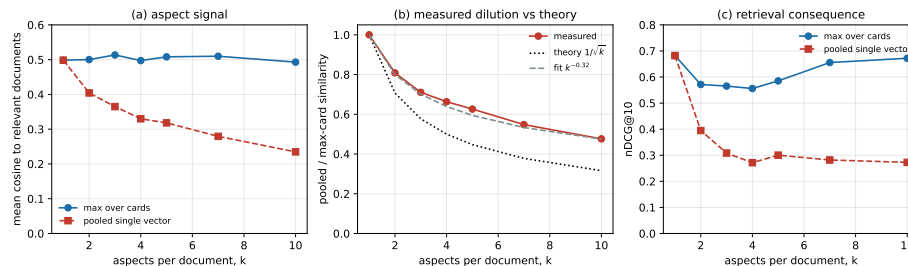


Fig. 1. Aspect dilution as a function of k . Left: the aspect signal is preserved by cards and lost by pooling. Centre: measured dilution against the $1/\sqrt{k}$ prediction and a fitted power law. Right: the retrieval consequence.

Table 2. LIMIT-small. Deltas are against the pooled dense baseline, paired permutation with Holm correction over 1,000 queries. Chunking and carding both raise the embedding count from one per document to many; only carding recovers the lost performance.

system	nDCG@10	R@2	R@10	Δ vs pooled p
BM25	0.997	0.991	1.000	+0.683 < 0.001
dense, pooled	0.314	0.162	0.480	baseline
dense, 30-word chunks	0.390	0.225	0.549	+0.076 < 0.001
dense, per-aspect cards	0.988	0.965	0.997	+0.674 < 0.001
cards $\oplus$ BM25 (RRF)	0.997	0.990	1.000	+0.683 < 0.001

4 Capacity: does alignment or count explain the gain?

The previous section compares cards against a single pooled vector, which leaves an obvious objection open: cards store more vectors per document, so perhaps capacity alone explains the gain and any decomposition would do. This section answers that objection with a control.

Setting. LIMIT [1] is constructed to expose the representational ceiling of single-vector retrieval. Each document names a person and lists the things they like; each query asks who likes one specific thing. The small variant has 46 documents, 44.0 attributes each, and 1,000 queries, so the task looks trivial and yet defeats dense retrievers. It also hands us an unambiguous aspect decomposition, which makes it the sharpest available test of whether alignment matters. We compare five systems, all scored at document level: BM25; a pooled dense vector; the same text cut into 30-word windows with 10-word overlap (186 chunks); the same text cut into per-attribute cards (2,024 cards); and reciprocal rank fusion of the card system with BM25.

Alignment, not count. Table 2 separates the two explanations. Chunking multiplies the embedding budget and recovers 0.076 nDCG@10. Carding multiplies

the same budget and recovers 0.674, roughly nine times as much, from identical source text under an identical encoder. The difference between the two is only where the boundaries fall: a 30-word window spans several attributes and dilutes each of them in the same way the pooled vector does, whereas a card corresponds to exactly the unit a query names. Capacity is necessary and not sufficient; the decomposition has to match the query distribution.

What this result does not show. LIMIT supplies a perfect decomposition for free, and its queries are effectively exact-attribute lookups, which is why BM25 wins outright and the hybrid only matches it. Real corpora arrive undecomposed, the taxonomy must be designed and may be misaligned, and the honest reading of this table is as an upper bound on what alignment can buy rather than as an estimate of what it will buy. We also note the cost: 44 cards per document here, far above the small constant that the pattern implies elsewhere, so the storage multiplier is a property of the corpus and must be reported per corpus rather than assumed.

5 Limitations

Three limitations bound what has been shown so far.

First, both results here are on constructed corpora. The synthetic corpus fixes k by construction, and LIMIT is adversarial by design. Neither tells us what happens on prose written by people for their own purposes, which is the setting the pattern is meant for.

Second, the aspect decomposition is given rather than inferred in both cases. The practical question is whether a taxonomy designed in advance, and applied by an imperfect extractor, retains a useful share of the benefit.

Third, a single encoder is used throughout. Anisotropy differs between encoders, and the dilution exponent is likely to move with it.

6 Reproducibility

Every number in this paper is produced by one command per experiment against committed code and seeds, and every per-query score is released alongside the aggregates so that the significance tests can be recomputed independently.

Determinism is enforced rather than assumed: two seeded runs must produce byte-identical per-query output, which is checked in the repository. Reaching that required accumulating similarity in double precision, because on some platforms a single-precision matrix product varies by around 10^{-8} between runs with the reduction order, enough to move a recorded value across a rounding boundary.

Metric implementations are covered by hand-worked golden tests, and the selection policies by property tests, for example that maximal marginal relevance with $\lambda = 1$ reduces exactly to relevance ranking and that reciprocal rank fusion is invariant to the order of its input lists.

References

1. Weller, O., Boratko, M., Naim, I., Lee, J.: On the theoretical limitations of embedding-based retrieval. arXiv preprint arXiv:2508.21038 (2025)